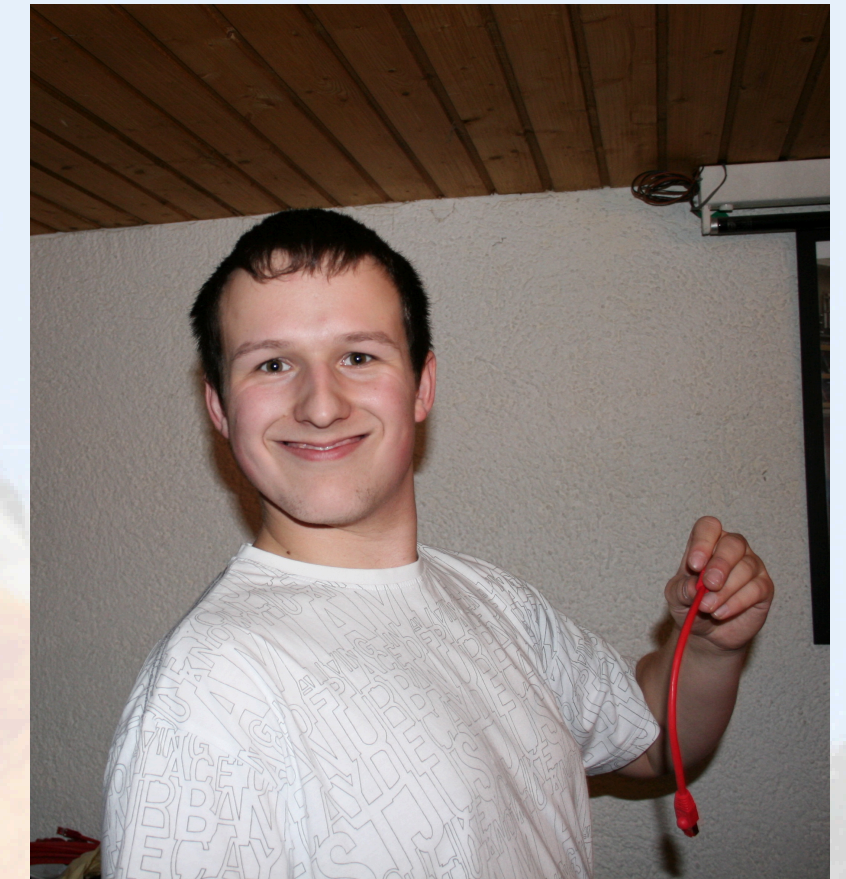


Foreman + OpenBolt

a <3 story in three acts

\$ whoami

- Tim 'bastelfreak' Meusel
- Puppet Contributor since 2012
- Merging stuff on [Vox Pupuli](#) since 2015
- Vox Pupuli PMC member
- Principal IT Automation Consultant at [betadots](#)









OpenBolt

- Runs tasks via ssh or WinRM on remote systems
 - CLI only
- A task is a executable binary/script + json file
- Also supports Podman/Docker/Jails/lxd targets



OpenBolt

```
{  
  "puppet_task_version": 1,  
  "supports_noop": false,  
  "description": "Switch feat branch to production",  
  "parameters": {  
    "control_repo_branch": {  
      "description": "Control-repo branch",  
      "type": "String[1]"  
    }  
  }  
}
```

```
#!/bin/bash
```

```
set -euo pipefail
```

```
pushd /tmp/control-repo || exit  
git branch --move production old_prod  
git branch --move "$PT_control_repo_branch" production
```



OpenBolt

- Runs tasks via ssh or WinRM on remote systems
 - CLI only
- A task is a executable binary/script + json file
- Also supports Podman/Docker/Jails/lxd targets
- Input and output of each task is JSON
 - makes parsing, scripting and concatenating easy



Convert the PE console into a foreman plugin #28

 Open

bastelfreak opened on Apr 16, 2024



Use Case

Foreman is an established open source tool for life cycle management. It has a working open source community, different companies offer commercial support/training/consulting for it. There are also commercial downstream products like RedHat Satellite or [Orcharhino](#). Foreman is used for many on premise setups, it has a proper CI setup and a plugin setup. It already has support to view puppet reports and facts. With [HDM](#) it has also an option to view, edit and diff Hiera data. Foreman can be used as an ENC and as hiera backend. And it has support for run remote jobs.

Foreman and the PE console overlap a lot in their feature set. I think Foreman is the better implementation because it looks more healthy. There are more docs, in general more activity and people can contribute to it. Contribute to the PE console is impossible and feature requests are mostly ignored.

I think it makes sense to replace the standalone PE console with a set of foreman plugins. Even if those are commercial/closed source people can at least contribute fixes to the rest of foreman and I assume that maintenance of the whole PE console codebase requires more time than it takes to maintain a small set of foreman plugins.

And since many setups already use Satellite/Orcharhino/Foreman they are already used to the UI and they don't need a second Web UI.

github.com/puppetlabs/puppet-enterprise_issues/issues/28

How it started

RFC: Integrating bolt into remote execution

■ Development ■ RFCs puppet remoteexecution rfc



bastelfreak

1 Oct 2024

Hi!

I would like to add support for bolt tasks to the remote execution architecture in foreman.

What's bolt?

An open source tool from same people as Puppet ([Bolt Documentation](#) ¹⁰). It supports running commands (shell or powershell), scripts (anything that's executable), tasks or plans.

What's a task?

A task is anything that's executable + a json file describing the purpose of a task and optional input parameters with datatypes. Tasks return json. bolt has a CLI interface and an API, no web UI. bolt can use different transport protocols, ssh and winrm are available by default. It has a plugin system.

Rationale:

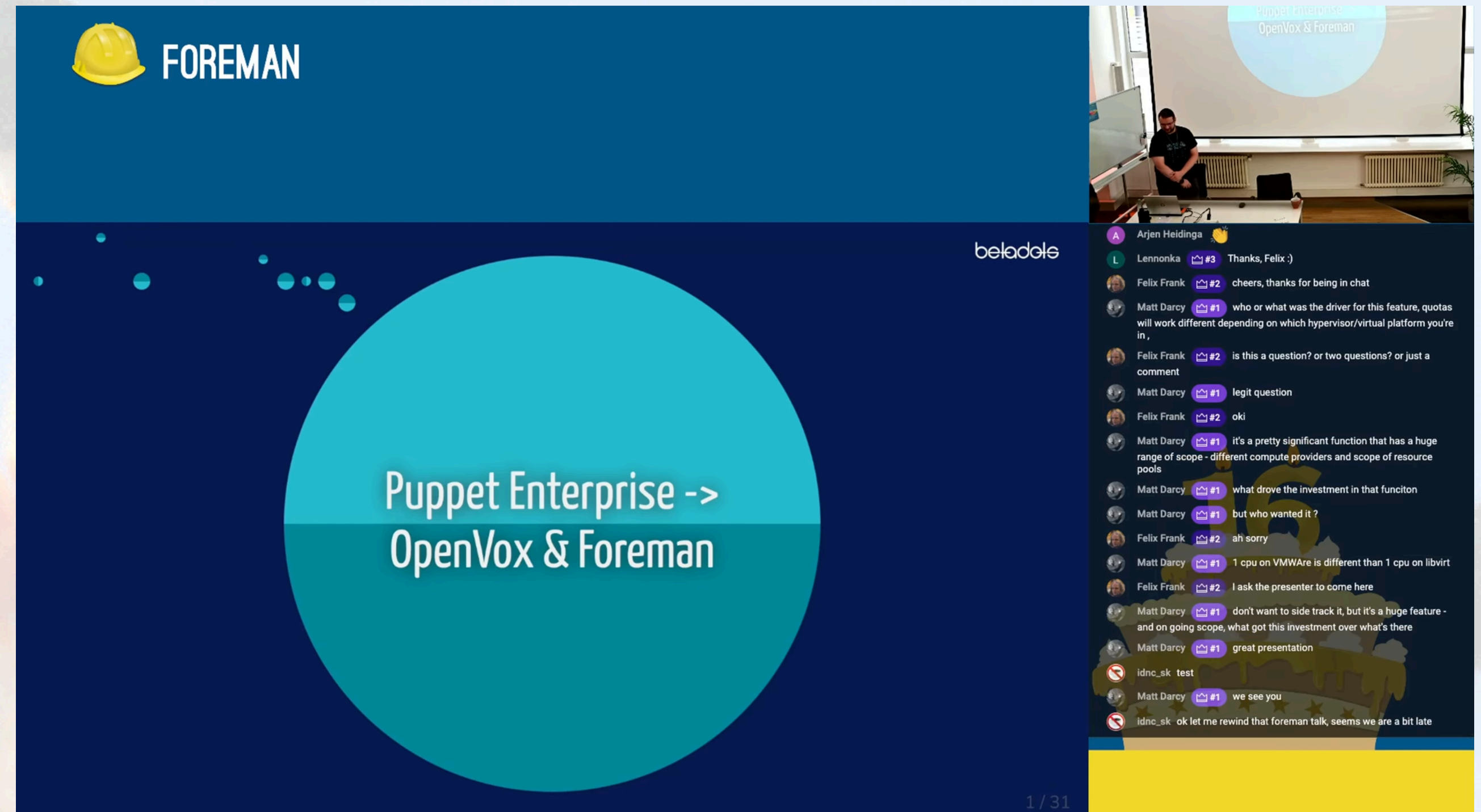
Puppet Enterprise (costs per node per year) has a Web UI to start tasks. I think it would be a great fit for the foreman UI to also support bolt tasks. bolt tasks are heavily used within the Puppet Enterprise and Puppet Opensource ecosystem for orchestration. For example OS patching whole clusters or doing software rollouts.

community.theforeman.org/t/rfc-integrating-bolt-into-remote-execution

OpenBolt

How it started

Act 1



The screenshot shows a presentation slide with a dark blue header containing a yellow hard hat icon and the word "FOREMAN". The main content area features a large teal circle with the text "Puppet Enterprise -> OpenVox & Foreman". To the right of the slide is a chat window with a dark background and the "beladots" logo. The chat contains a list of messages from participants like Arjen Heidinga, Lennonka, Felix Frank, and Matt Darcy, discussing technical details about Puppet Enterprise, OpenVox, and Foreman. A small video inset in the top right corner shows a person standing next to a screen displaying the same presentation slide. The bottom right corner of the slide shows the page number "1 / 31".

FOREMAN

Puppet Enterprise ->
OpenVox & Foreman

beladots

Arjen Heidinga 🍌
Lennonka 🏠 #3 Thanks, Felix :)
Felix Frank 🏠 #2 cheers, thanks for being in chat
Matt Darcy 🏠 #1 who or what was the driver for this feature, quotas will work different depending on which hypervisor/virtual platform you're in,
Felix Frank 🏠 #2 is this a question? or two questions? or just a comment
Matt Darcy 🏠 #1 legit question
Felix Frank 🏠 #2 oki
Matt Darcy 🏠 #1 it's a pretty significant function that has a huge range of scope - different compute providers and scope of resource pools
Matt Darcy 🏠 #1 what drove the investment in that function
Matt Darcy 🏠 #1 but who wanted it?
Felix Frank 🏠 #2 ah sorry
Matt Darcy 🏠 #1 1 cpu on VMWare is different than 1 cpu on libvirt
Felix Frank 🏠 #2 I ask the presenter to come here
Matt Darcy 🏠 #1 don't want to side track it, but it's a huge feature - and on going scope, what got this investment over what's there
Matt Darcy 🏠 #1 great presentation
idnc_sk test
Matt Darcy 🏠 #1 we see you
idnc_sk ok let me rewind that foreman talk, seems we are a bit late

1 / 31

- Recording: youtu.be/NRtstDVtpyk?t=21810



OpenBolt

How it started

Act 1

Act 2



bastelfreak

Jan 30

Hey people, I will also do a live demo at the Config Management camp in Ghent next week:

 cfp.cfgmgmtcamp.org



Bringing OpenBolt into Foreman CfgMgmtCamp 2026 Ghent ⁷

[OpenBolt](<https://github.com/OpenVoxProject/openbolt>) is a CLI application from the [OpenVoxProject](<https://voxpupuli.org/openvox/>). It provides different ways to run your beloved sysadmin scripts with proper guardrails and it makes it easy to...

Hit me up if you have questions or thoughts!

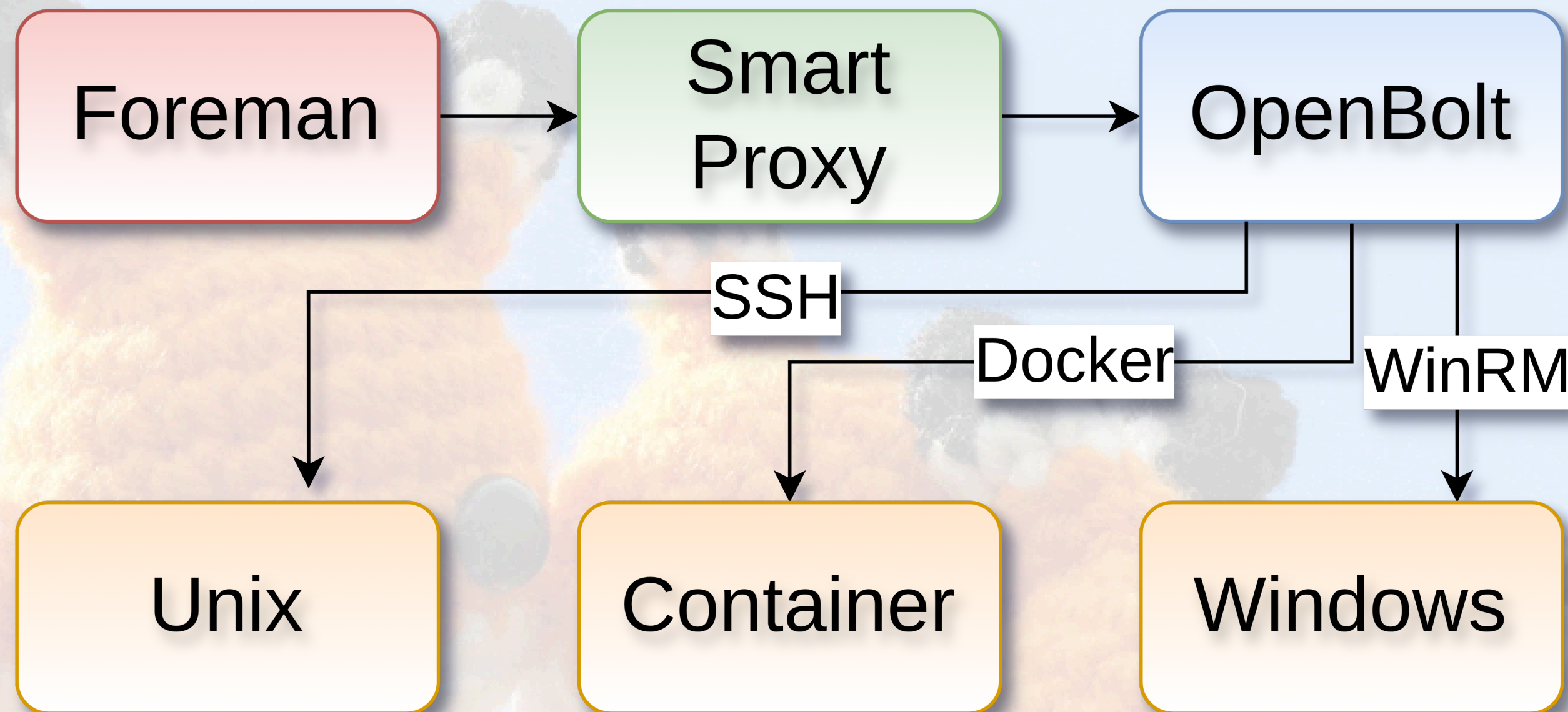
- Slideset: bastelfreak.de/cfgmgmtcamp2026/openbolt.html

OpenBolt

How it started

Act 1

Act 2



OpenBolt

How it started

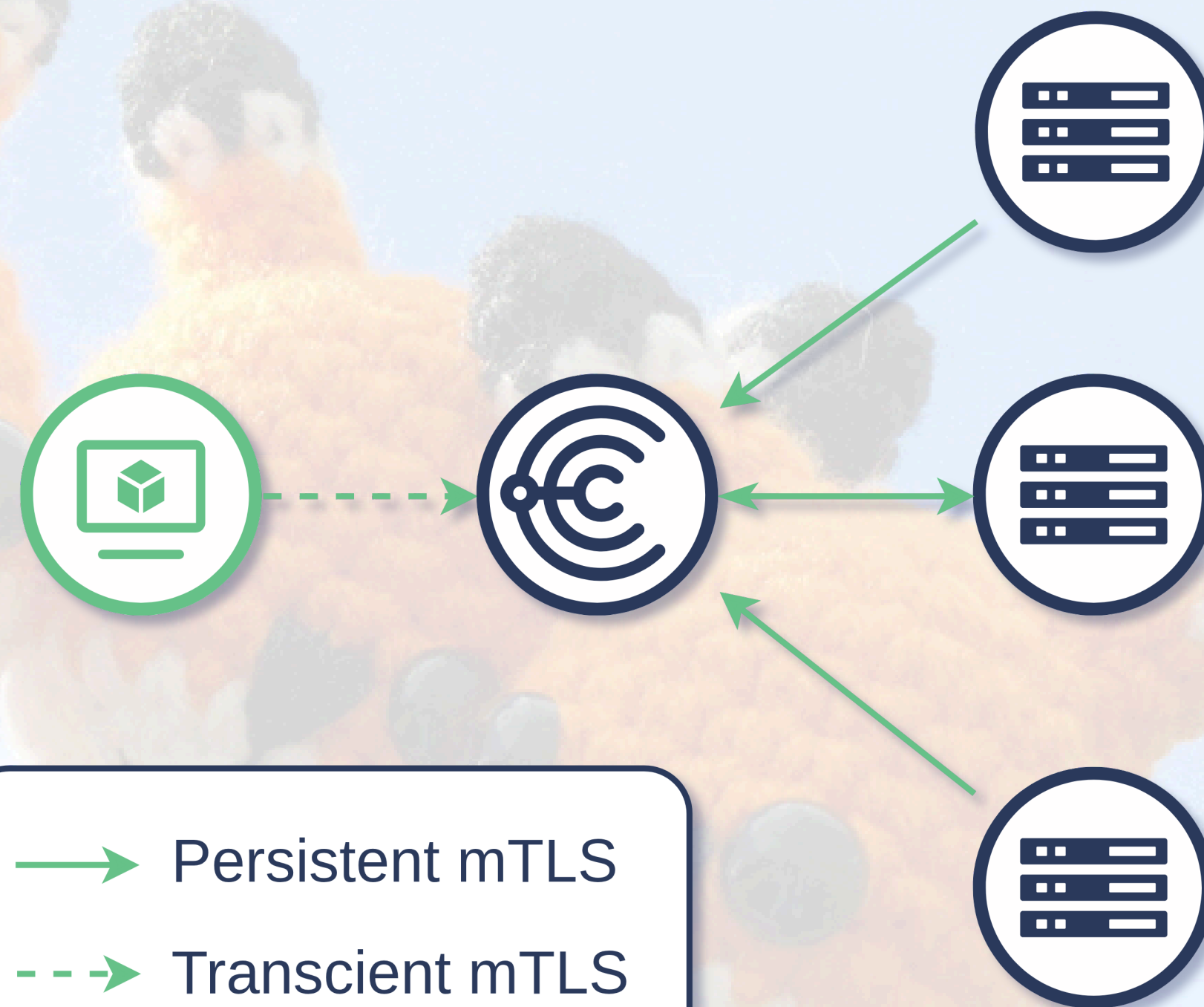
Act 1

Act 2

Client

Broker

Server



- Choria Orchestration System

OpenBolt

How it started

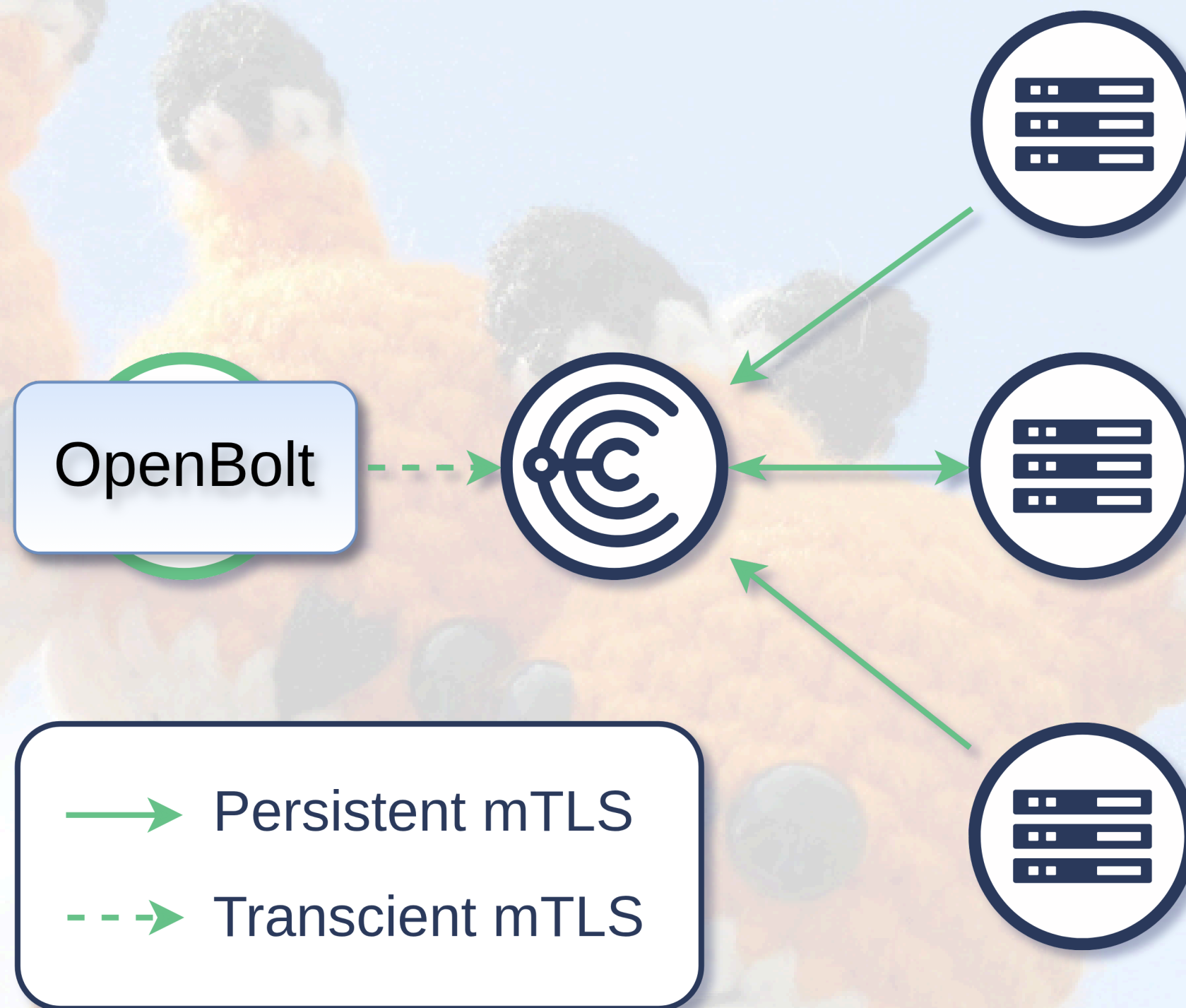
Act 1

Act 2

Client

Broker

Server



- Choria Orchestration System

Act 3

OpenBolt

- github.com/overlookinfra/foreman_openbolt - OpenBolt Plugin for Foreman

How it started

Act 1

Act 2

Act 3



OpenBolt

How it started

Act 1

Act 2

Act 3

- github.com/overlookinfra/foreman_openbolt - OpenBolt Plugin for Foreman
- github.com/overlookinfra/smart_proxy_openbolt - OpenBolt Smartproxy Plugin



OpenBolt

How it started

Act 1

Act 2

Act 3

- github.com/overlookinfra/foreman_openbolt - OpenBolt Plugin for Foreman
- github.com/overlookinfra/smart_proxy_openbolt - OpenBolt Smartproxy Plugin
- github.com/theforeman/foreman-installer/pull/1049 - foreman-installer Integration



OpenBolt

How it started

Act 1

Act 2

Act 3

- github.com/overlookinfra/foreman_openbolt - OpenBolt Plugin for Foreman
- github.com/overlookinfra/smart_proxy_openbolt - OpenBolt Smartproxy Plugin
- github.com/theforeman/foreman-installer/pull/1049 - foreman-installer Integration
- github.com/OpenVoxProject/openbolt/pull/206 - Integrate Choria Transport into OpenBolt



OpenBolt

How it started

Act 1

Act 2

Act 3

- github.com/overlookinfra/foreman_openbolt - OpenBolt Plugin for Foreman
- github.com/overlookinfra/smart_proxy_openbolt - OpenBolt Smartproxy Plugin
- github.com/theforeman/foreman-installer/pull/1049 - foreman-installer Integration
- github.com/OpenVoxProject/openbolt/pull/206 - Integrate Choria Transport into OpenBolt
- All changes are merged & released & packaged for Foreman 3.17.1+



Part 3 of 4?!

